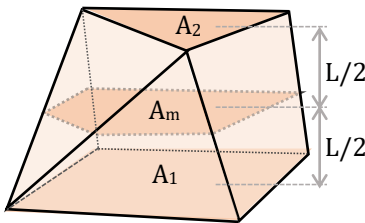




## Prismatoid

A prismatoid is a polyhedron having for bases two polygons in parallel planes and for lateral faces triangles or trapezoids with one side lying in one base, and the opposite vertex or side lying in the other base of the polyhedron.



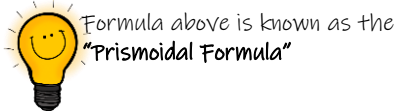
$$V = \frac{L}{6} (A_1 + 4A_m + A_2)$$

Where: Excel Review Center

$A_1$  and  $A_2$  = end areas

$A_m$  = area at midsection

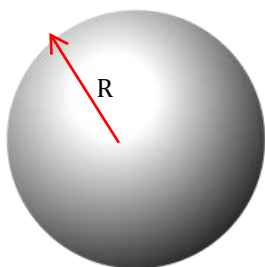
$L$  = distance between end areas



Formula above is known as the "Prismoidal Formula"

## Sphere

A sphere is a solid bounded by a closed surface every point of which is equidistant from a fixed point called center.

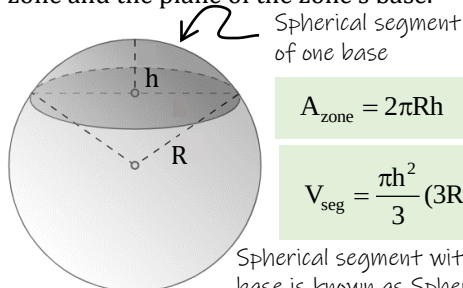


$$V = \frac{4}{3} \pi R^3$$

$$A_s = 4\pi R^2$$

**Spherical zone** is that portion of the surface of a sphere included between two parallel planes.

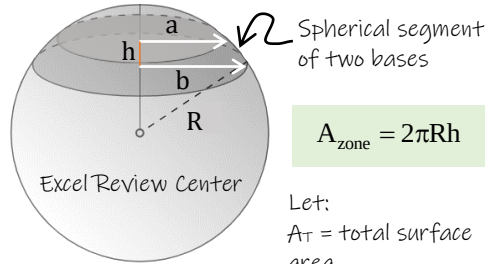
**Spherical segment** is a solid bounded by a zone and the plane of the zone's base.



$$A_{\text{zone}} = 2\pi Rh$$

$$V_{\text{seg}} = \frac{\pi h^2}{3} (3R - h)$$

Spherical segment with one base is known as Spherical Cap.



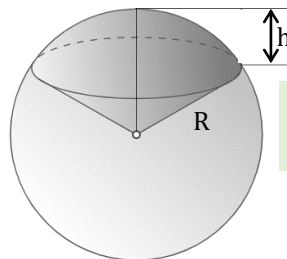
$$A_{\text{zone}} = 2\pi Rh$$

Let:  
 $A_T$  = total surface area

$$A_T = \pi (2Rh + a^2 + b^2)$$

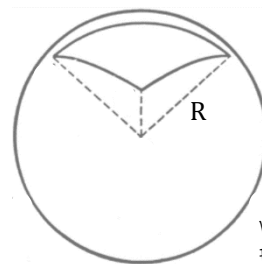
$$V_{\text{seg}} = \frac{\pi h}{6} (3a^2 + 3b^2 + h^2)$$

**Spherical sector** is a solid generated by rotating a sector of a circle about an axis which passes through the center of the circle by which contains no point inside the sector.



$$V = \frac{1}{3} A_{\text{zone}} R$$

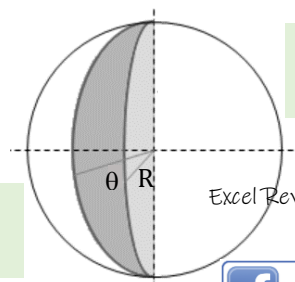
**Spherical pyramid** is a pyramid formed by a portion of a surface of a sphere as base and whose elements are the edges from the vertices of the base to the center of the sphere.



$$V = \frac{\pi R^3 E}{540}$$

where:  
 $E$  = spherical excess

**Spherical wedge** is a portion of a sphere bounded by two half great circles and an included arc.



$$V = \frac{\pi R^3 \theta}{270}$$

Excel Review Center

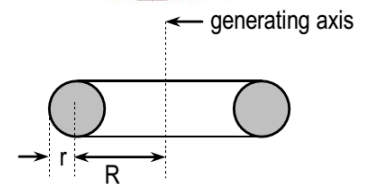
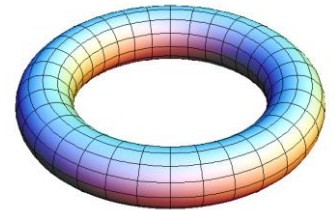


Excel Review Center

## Torus

Excel Review Center

A torus is a solid formed by revolving a circle about a line not intersecting it.



$$V_{\text{torus}} = 2\pi^2 R r^2$$

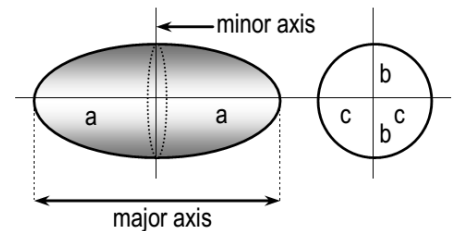
$$A_L = 4\pi^2 R r$$

$A_L$  = lateral area of torus

## Ellipsoid

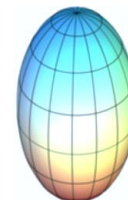
Excel Review Center

An ellipsoid is a solid formed by revolving the ellipse about its axis. It is also known as Spheroid.



$$V = \frac{4}{3} \pi abc$$

**Prolate spheroid** is a solid formed by revolving an ellipse about its major axis.



$$V = \frac{4}{3} \pi a b^2$$

**Oblate spheroid** is a solid formed by revolving an ellipse about its minor axis.

$$V = \frac{4}{3} \pi a^2 b$$

